

(1.1) Dokažte, že v libovolném komplexním vektorovém prostoru V se skalárním součinem $\langle, \rangle$ platí (pro libovolné $u, v \in V$)

$$\operatorname{Re}(\langle u, v \rangle) = \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2),$$

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4}(\|u - iv\|^2 - \|u + iv\|^2),$$

kde $\operatorname{Re}(x)$ značí reálnou a $\operatorname{Im}(x)$ imaginární část komplexního čísla x . (Pro inspiraci vizte tvrzení 8.32 ve skriptech.)

a) $\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)$

Z Definice 8.14 (SSS) a (SL2) dostáváme

$$\|\vec{u} + \vec{v}\|^2 = \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle = \|\vec{u}\|^2 + \langle \vec{u}, \vec{v} \rangle + \overline{\langle \vec{u}, \vec{v} \rangle} + \|\vec{v}\|^2$$

$$\begin{aligned} \|\vec{u} - \vec{v}\|^2 &= \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, -\vec{v} \rangle + \langle -\vec{v}, \vec{u} \rangle + \langle -\vec{v}, -\vec{v} \rangle = \|\vec{u}\|^2 - \langle \vec{u}, \vec{v} \rangle + (-1) \overline{\langle \vec{u}, \vec{v} \rangle} (-1) \cdot (-1) \|\vec{v}\|^2 = \\ &= \|\vec{u}\|^2 - \langle \vec{u}, \vec{v} \rangle - \overline{\langle \vec{u}, \vec{v} \rangle} + \|\vec{v}\|^2 \end{aligned}$$

Potom

$$\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2 = \|\vec{u}\|^2 + \langle \vec{u}, \vec{v} \rangle + \overline{\langle \vec{u}, \vec{v} \rangle} + \|\vec{v}\|^2 - \|\vec{u}\|^2 + \langle \vec{u}, \vec{v} \rangle + \overline{\langle \vec{u}, \vec{v} \rangle} - \|\vec{v}\|^2 = 2(\langle \vec{u}, \vec{v} \rangle + \overline{\langle \vec{u}, \vec{v} \rangle})$$

Protože $x + \bar{x} = 2\operatorname{Re}(x)$, platí

$$2(\langle \vec{u}, \vec{v} \rangle + \overline{\langle \vec{u}, \vec{v} \rangle}) = 2 \cdot 2\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = 4\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)$$

Odtud plyne, že

$$\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4}(\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)$$

b) $\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle)$

Opět z Definice 8.14 (SSS) a (SL2) dostáváme

$$\begin{aligned} \|\vec{u} + i\vec{v}\|^2 &= \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, i\vec{v} \rangle + \langle i\vec{v}, \vec{u} \rangle + \langle i\vec{v}, i\vec{v} \rangle = \|\vec{u}\|^2 + i\langle \vec{u}, \vec{v} \rangle + \overline{i\langle \vec{u}, \vec{v} \rangle} + i \cdot \bar{i} \|\vec{v}\|^2 = \\ &= \|\vec{u}\|^2 + i\langle \vec{u}, \vec{v} \rangle - i\overline{\langle \vec{u}, \vec{v} \rangle} + \|\vec{v}\|^2 \end{aligned}$$

$$\begin{aligned} \|\vec{u} - i\vec{v}\|^2 &= \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, -i\vec{v} \rangle + \langle -i\vec{v}, \vec{u} \rangle + \langle -i\vec{v}, -i\vec{v} \rangle = \|\vec{u}\|^2 - i\langle \vec{u}, \vec{v} \rangle + \overline{-i\langle \vec{u}, \vec{v} \rangle} + (-i) \cdot \overline{(-i)} \|\vec{v}\|^2 = \\ &= \|\vec{u}\|^2 - i\langle \vec{u}, \vec{v} \rangle + i\overline{\langle \vec{u}, \vec{v} \rangle} + \|\vec{v}\|^2 \end{aligned}$$

$$\begin{aligned} \|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2 &= \|\vec{u}\|^2 - i\langle \vec{u}, \vec{v} \rangle + i\overline{\langle \vec{u}, \vec{v} \rangle} + \|\vec{v}\|^2 - \|\vec{u}\|^2 - i\langle \vec{u}, \vec{v} \rangle + i\overline{\langle \vec{u}, \vec{v} \rangle} - \|\vec{v}\|^2 = \\ &= -2i(\langle \vec{u}, \vec{v} \rangle - \overline{\langle \vec{u}, \vec{v} \rangle}) \end{aligned}$$

Protože $x - \bar{x} = 2i \operatorname{Im}(x)$, dostáváme

$$-2i(\langle \vec{u}, \vec{v} \rangle - \overline{\langle \vec{u}, \vec{v} \rangle}) = -2i \cdot 2i \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = 4 \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle)$$

Odtud plyne, že

$$\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4}(\langle \vec{u} - i\vec{v}, \vec{u} + i\vec{v} \rangle)$$